MAKEGOOD POWER CHAIR

Configurator Guide

Using and extending the build configurator

R.O.A.R. / TSSFAA with **MakeGood INC**
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Private pre-release draft. Not yet safe to build from. Prices are September 2026 US retail.

`configurator/index.html` is a single self-contained web page — no build step, no server, no dependencies beyond a font. Open it in any browser. It walks a builder through 18 decisions and produces a **build sheet**: parts list with cost ranges, warnings, and a verification checklist keyed to requirement IDs.

Worked examples for seven configurations are in `docs/examples/`; their totals are compared in `price-matrix.md`.

Using it

1. Open the page. Defaults load (configuration B in the price matrix).
2. Work top to bottom. Options that conflict with an earlier choice are greyed out with the reason on hover — e.g. choose wheelchair gearmotors and the VESC/ODrive options disable with "Brushed motors only".
3. Watch the summary bar: **"Build sheet: ready to review"** or **"not ready"** with the count of blocking and caution warnings. A blocking warning highlights the fieldset that caused it.
4. **Copy as Markdown** exports the full build sheet (what the files in `docs/examples/` are). **Copy JSON** exports the configuration for re-import or for attaching to a GitHub issue.
5. Your choices persist in the browser (localStorage). **Reset** returns to defaults.

The 18 decisions

#	Decision	Req.	Options (key → label)
1	Driver mass (max)	PAS-1	`15` up to 15 kg (ages ~1–3) · `25` up to 25 kg (ages ~3–8)
2	Friend riding along	§8	`none` · `trailer` · `seat` buddy seat · `board` standing board · `play` shared-play modes only
3	Manual / power mechanism	§9	`lift` lever-deployed drive wheels · `pod` removable power pod · `friction` friction drive · `poweronly` (blocked)
4	Drive motors	DRV-2	`hub36` hoverboard-class hub · `wcbrushed` wheelchair gearmotor · `wcbldc` wheelchair brushless hub wheel
5	Motor source	DRV-2a	`new` · `donated` (hoverboard only) · `salv` salvaged · `mixed`
6	Drive wheel size	DRV-3	`6.5` · `8` · `10` in
7	Motor controller	ELE-2b	`vesc` · `odrive` · `zs` (bench only) · `reflash` · `roboclaw` · `sabertooth` · `wcctrl` standard wheelchair controller
8	Battery	§6	36 V: `2ah` · `5ah` · `hot` 4×5 Ah hot-swap — 24 V: `b24_6` · `b24_10` · `b24hot` · `sla`
9	Regenerative braking	SAF-10	`off` · `on`

#	Decision	Req.	Options (key → label)
10	Parking brake (multi)	SAF-1/2	`integ` motor's EM brake · `manual` wheel locks · `pin` solenoid pin · `em` add-on EM · `cable` disc caliper
11	Assistive driving computer	4.7	`none` · `ventuno` · `unoq` · `other`
12	Assist features (multi)	ASD	`collide` · `drop` · `street` · `tip` · `power`
13	Assist default level	ASD-2	`warn` · `slow` · `stop` (with override)
14	Ventilator	§7	`none` · `indep` own batteries · `fed` from chair
15	Driver input (multi)	INP-2	`joy` · `sw` switches · `scan` single-switch scanning · `snp` Sip-N-Puff · `head` head array
16	Supervision (multi)	4.6.2	`deadman` · `estop` · `takeover` · `tether` · `enable` · `timed` · `prox` · `indep`
17	Attendant link	ATT-4	`ble` · `rc` 2.4 GHz with failsafe · `none`
18	Parent settings interface	CFG-3	`key` on-chair key + display · `app` phone · `both`

How the rules work

Everything lives in one `<script>` block. Three structures matter:

`P` — the price book. `key: [low, high, basis]`. `null` means TBD. Rendered in `price-book.md`.

`DECISIONS` — the form. Each option is `[key, label, sublabel, availableFn, disabledReason]`. `availableFn(state)` returns false to grey the option out. Example: ODrive is available only when `s.drive !== "wcbrushed"`.

`compute()` — the rules. Reads `state`, calls three helpers: - `add(category, item, qty, note, priceKey)` — a BOM line - `warn("stop" | "caution" | "info", text, fieldsetId)` — `stop` blocks the build and highlights the fieldset - `C.push(text)` — a verification checklist item

Rules are grouped by subsystem in source order: always-included parts, drive, controller, battery, passenger, dual mode, brakes, assist, ventilator, inputs, supervision/link, config UI.

Blocking rules (as of rules 0.4)

Condition	Why
`poweronly` dual mode	Fails dual-mode requirement (1.5a, MEC-7)
ZS-X11H controller	Open-loop, no current limit, no watchdog — fails ELE-2
No parking brake	SAF-1 requires hold on 1:12 slope, power off
No driver input	Nothing to drive with

Condition	Why
Remote supervision option with no attendant link	Needs BLE or RC

Automatic derivations worth knowing

- Switching to a wheelchair motor family auto-adds `integ` (the motor's own EM brake) to the brake list.
- A dump resistor is added whenever regen is on, hot-swap is selected (ideal diodes block regen into packs), or ODrive is selected (headroom to the 50.5 V trip).
- Lift/pod dual mode doubles the wheel-lock count (printed wheels *and* drive wheels) unless the motors have integrated brakes.
- Choosing an assist computer adds a DC-DC, an override switch matched to the driver input, and an annunciator.
- `tip` and `power` assist features need no assist computer — they run on the supervisor.

Extending it

- **New part / price:** add to `P`, reference by key in `add(...)`. Mark it `est` until a listing is found.
- **New option:** append to the `opts` array of the decision; add an `availableFn` if it conflicts with anything; add rules in `compute()`.
- **New requirement ID:** cite it in the `note` of the BOM line or the checklist text so the build sheet stays traceable.
- Bump the `rules` version string in the JSON export (`"rules":"0.4"`) when behaviour changes, and regenerate `docs/examples/` and `price-matrix.md` with `docs/tools/run-scenarios.js`.

Regenerating the docs

```
node docs/tools/run-scenarios.js configurator/index.html   # → docs/examples/*.md,
docs/price-matrix.md
node docs/tools/pricebook.js                                # → docs/price-book.md
```

Both scripts run the configurator's own JavaScript in Node with a stub DOM, so the generated docs are exactly what the page would produce.

Limits

- Prices are a snapshot (Sept 2026, US). The tool does not fetch live prices.
- It does not know about your printer, filament, or labour.
- It does not replace the requirements doc — it enforces a subset of its rules.

- Rules are only as good as what we've encoded. If a combination you know is bad isn't flagged, open an issue.